Example 6.41

Find the vector and Cartesian equations of the plane passing through the point with position vector $4\hat{i}+2\hat{j}-3\hat{k}$ and normal to vector $2\hat{i}-\hat{j}+\hat{k}$.

Solution

If the position vector of the given point is $\vec{a}=4\hat{i}+2\hat{j}-3\hat{k}$ and $\vec{n}=2\hat{i}-\hat{j}+\hat{k}$, then the equation of the plane passing through a point and normal to a vector is given by $(\vec{r}-\vec{a})\cdot\vec{n}=0$ or $\vec{r}\cdot\vec{n}=\vec{a}\cdot\vec{n}$.

Substituting $\vec{a} = 4\hat{i}+2\hat{j}-3\hat{k}$ and $\vec{n}=2\hat{i}-\hat{j}+\hat{k}$ in the above equation, we get

$$\vec{r}\cdot(2\hat{i}-\hat{j}+\hat{k}) = (4\hat{i}+2\hat{j}-3\hat{k})\cdot(2\hat{i}-\hat{j}+\hat{k})$$

Thus, the required vector equation of the plane is $\vec{r}\cdot(2\hat{i}-\hat{j}+\hat{k})=3$. If $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$ then we get the Cartesian equation of the plane $2x-y+z=3$. ■

Example 6.42

A variable plane moves in such a way that the sum of the reciprocals of its intercepts on the coordinate axes is a constant. Show that the plane passes through a fixed point

Solution

The equation of the plane having intercepts a,b,c on the x,y,z axes respectively is $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$. Since the sum of the reciprocals of the intercepts on the coordinate axes is a constant, we have $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=k$, where k is a constant, and which can be written as $\frac{1}{a}\left(\frac{1}{k}\right)+\frac{1}{b}\left(\frac{1}{k}\right)+\frac{1}{c}\left(\frac{1}{k}\right)=1$.

This shows that the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ passes through the fixed point $\left(\frac{1}{k},\frac{1}{k},\frac{1}{k}\right)$. ■

EXERCISE 6.6

1. Find the vector equation of a plane which is at a distance of 7 units from the origin having $3,-4,5$ as direction ratios of a normal to it.
2. Find the direction cosines of the normal to the plane $12x+3y-4z=65$. Also, find the non-parametric form of vector equation of a plane and the length of the perpendicular to the plane from the origin.
3. Find the vector and Cartesian equations of the plane passing through the point with position vector $2\hat{i}+6\hat{j}+3\hat{k}$ and normal to the vector $\hat{i}+3\hat{j}+5\hat{k}$.
4. A plane passes through the point $(-1,1,2)$ and the normal to the plane of magnitude $3\sqrt{3}$ makes equal acute angles with the coordinate axes. Find the equation of the plane.
5. Find the intercepts cut off by the plane $\vec{r}\cdot(6\hat{i}+4\hat{j}-3\hat{k})=12$ on the coordinate axes.
6. If a plane meets the coordinate axes at A,B,C such that the centriod of the triangle ABC is the point (u,v,w), find the equation of the plane.

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